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MIL-C-70575B (AR)
26 FEBRUARY 199-2
SUPERSEDING
(See 6.6)

MILITARY SPECIFICATION

CIRCUIT BOARD, DISPLAY CONTROL PANEL: **9360884**

This specification is approved for use by the U. S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 scope. This specification covers the requirements, examinations and tests for the Display Control Panel Built-In-Test (BIT) circuit card assembly, hereinafter referred to as the Display Control Panel (DCP) BIT assembly, used on the Vulcan Air Defense System, Anti-Aircraft, Self-Propelled (M163A2), and Towed (M167A2) Vehicles (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 specifications, standards and handbooks The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2) .

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for

MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

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FSC 1285

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STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 RRobbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6-4)

U. S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9360884 - Display Control Panel Assembly
9360886 - Display Control Schematic Diagram
9361257 - Test Plug Jll
9361343 - Test Program Set
9397373 - Interface Device Card C
9397501 - Cable IDC WI
12950877 - Requirements for Soldering Electrical and Electronic Assemblies

SPECIAL PACKAGING INSTRUCTION

PS 9360884 - packaging of Display Control Panel Circuit Board

(Copies of drawings, publications and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publication. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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INSTITUTE FOR INTERCONNECTING AND PACKAGING ELECTRONIC CIRCUITS

ANSI/IPC-DW-425 - Design and End Product Requirements for Discrete Wiring Boards

ELECTRONIC INDUSTRIES ASSOCIATION

EIA RS-422 - Electrical Characteristics of Balanced Voltage Digital Interface Circuits

statement standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies The components and assemblies shall comply with all requirements specified on Drawing 9360884, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the Display Control Panel (DCP) BIT assembly shall meet the requirements of MIL-F-13926 and ANSI/IPC-DW-425 as follows:

3.2.1 MIL-F-13926.

Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.

3.2.2 ANSI/IPC-DW-425.

a. ANSI/IPC-DW-425/11 Type Cl plated through connection.

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3.3 Environmental service conditions.

3.3.1 Vibration The DCP BIT assembly shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
5 to 15 hertz	0.06 inches.double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 55 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decade.
The 5-55-5 hertz sweep time is 7 minutes 35 seconds ± 10%.

3.3.2 shock. The DCP BIT assembly shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude.

3.3.3.1 Operating. The DCP BIT assembly shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg) .

3.3.3.2 Non-operating. The DCP BIT assembly shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

3.3.4 Temperature .

3.3.4.1 Operating The DCP BIT assembly shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40 deg F and +155 deg F.

3.3.4.2 Non-Operating (storage The DCP BIT assembly shall not suffer permanent damage when subjected to the constant temperature extremes of -60 deg F for a period of 24 hours and +160 deg F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating) The DCP BIT assembly shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

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3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram (9360886) of the DC? .

3.5 Performance.

3.5.1 Functional description. The DCP BIT assembly shall interface the Ballistics/BIT Panel to the Fire Control Processor (FCP). The DCP BIT assembly shall contain a UART link to transmit data to and receive data from the FCP. Differential line drivers and line receivers shall be used at both ends to ensure proper transfer of data. The DCP BIT assembly's UART shall convert received serial data to parallel format and convert parallel data to serial format for transmission to the FCP. A master reset of the UART shall be generated manually or by system power-up. The DCP BIT assembly has three major functions:

Transmit the status of Ballistics/BIT Panel switches S9 through S12, and S14 to FCP;

b. Conduct an End-Around-Test (EAT) of the DCP BIT assemblY UART; and

c. Convert binary data received from the FCP to alpha numeric data for display to the operator on the external digital value display of the Ballistics/BIT Panel.

The DCP BIT assembly shall contain circuitry to enable a lamp test of all segments of the digital value display and also contain resistor pull-up networks to maintain signal levels.

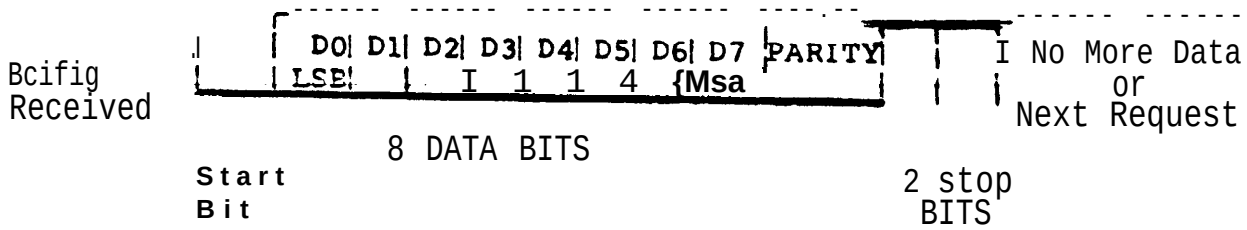
3.5.2 Power Requirements. The power requirements for the DCP BIT assembly shall be +5 Vdc ± 0.25 Vdc @ 2.5 amperes, with source impedance less than 0.050 ohms and DC ripple less than 0.1 volts peak to peak. Connectors are as follows:

Connectors: J1-35 = +5 Vdc
J1-59, 69, 70 = + 5 V RTN

3.5.3 Clock requirements. The clock input to the DCP BIT assembly applied at J1-3 (HI) "and J1-4 (LO) shall be differential with RS-422 voltage levels per 3.5.13. The clock frequency shall be 500 kHz ± 74 Hz. With power applied to the DCP BIT assembly per 3.5.2, the clock shall have TTL voltage levels per 3.5.12 at J1-27 (CLK) test point except that a pull-up resistor shall ensure the 3.0 minimum-CMOS threshold required by the UART.

3.5.4 UART receiver input data format. The input data to the DCP BIT assembly shall be a different signal with RS-422 voltage levels per 3.5.13 output at J1-1 (HI) and J1-2 (LO). With power applied no the DCF BIT assembly per 3.5.2, the input data shall have TTL voltage levels per 3.5.12 at J1-26 (RCV) test point except that a pull-up resistor shall insure the 3.0 V minimum CMOS threshold required by the UART. The input data format shall be per Figure (Receiver input data format).

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NOTES :

1. The data line is always high when no input data is being received.

2* The start bit is always low and the 2 stop bits are always high.

3. LSB = Least Significant BIT and is the first data bit received in each word.

   MSB = Most Significant BIT and is the last data bit received in each word.

4. The parity bit is a “don’t -care” even though the UART is wired to check odd parity, because the parity error flag output of the UART is not connected.

5. When a control word has been received by the UART, it is output on the UART’S Receiver Buffer Register (RBR) lines as follows:

   BIT DO is on RBR1

   BIT D1 is on RBR2

   .
   .
   .

   BIT D7 is on RBR8

FIGURE 1. Receiver input data format.

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3.5.5 UART transmitter output data format. The data transmitted from the DCP BIT assembly shall be a differential signal with RS-422 levels per 3.5.13 output at J1-5 (HI) and J1-6 (LO). With power applied to the DCP BIT assembly per 3.5.2, the transmitted output data shall have CMOS voltage levels per 3.5.14 at J1-28 (TRO) test point. The transmitted output data format shall be per Figure 2 (Transmitter output data format).

3.5.6 Master reset circuit.

3.5.6.1 Power-up reset. Upon power-up per 3.5.2 and 3.5.3, UART Master Reset signal $\overline{MR}$ (J1-67) shall remain low for 230 ± 113 microseconds and then go high.

3.5.6.2 $\overline{RESET}$ terminal. With power and clock applied as in 3.5.6.1, grounding $\overline{RESET}$ (J1-18) to +5 V RTN shall cause MR (J1-67) to go low for 8.0 ± 4.0 milliseconds then go high. (In the system, $\overline{RESET}$ is grounded when the Ballistics/BIT Panel FUNCTION switch, S14, is set to the OFF position.)

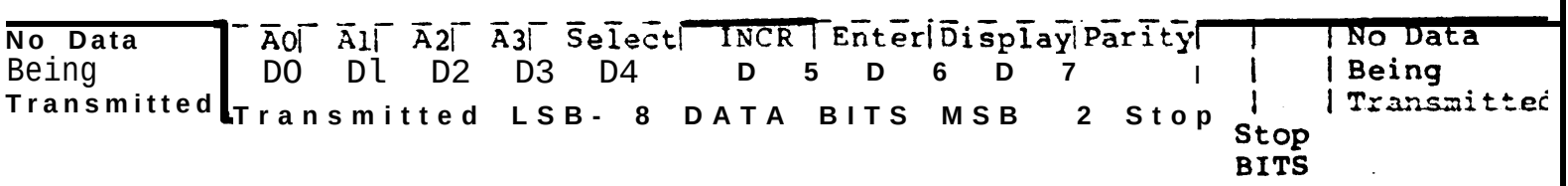
3.5.6.3 DCP BIT assembly reset. A low on $\overline{MR}$ (high on UART's MR line) shall cause DR (J1-31) to go low and TRO (J1-28) and DRR (J1-32) to go high, initializing the DCP BIT assembly for receiving data and inhibiting data transmission. After the first rising clock edge following MR going low, the following signals shall all be high:

a. $\overline{DECE}$ (J1-33)
b. D1 CLK (J1-24)
D2 CLK (J1-25)
d. D3 CLK (J1-34)
e. D4 (J1-68)
f. $\overline{TXDSEL}$ (J1-29)
g. TBRL (J1-30)

3.5.7 switch position data transmission.

3.5.7.1 Inialization. With power and clock applied per 3.5.2 and 3.5.3, respectively, the DCP BIT assembly shall reset per 3.5.6.

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NOTES :

1. The data line is always high when no data is being transmitted.

2. The start bit is always low and the 2 stop bits are always high.

3. LSB = Least Significant BIT and is the first data bit transmitted in each word.

   MSB = Most Significant BIT and is the last data bit transmitted in each word.

4. The parity bit is odd: the UART sums data bits DO through D7 and adds a one (makes the parity bit high) or adds a zero (makes the parity bit low) to cause the sum plus the parity bit to be an odd number.

5. Each data bit for transmission is input into the UART on the UART'S Transmitter Buffer Register (TBR) lines as follows:

Transmitter Buffer Register Line	MODE	
	Transmit Switch Positions	End-Around-Test
TBR1	A0	DO
TBR2	Al	D1
TBR3	A2	D2
TBR4	A3	D3
TBR5	SELECT	D4
TBR6	INCR	D5
TBR7	ENTER	D6
TBR8	DISPLAY	D7

FIGURE 2. Transmitter output data format.

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3.5.7.2 Control and data transition. When a complete control word with format per 3.5.4 and with data bits XXXXX01l (DO through D7, respectively) has been clocked into the UART (received), DR (J1-31) shall transition from low to high. (Bits marked "X" are "don't.-cares" and may be high or low.) Successive clock pulses shall cause the signal transitions shown in Figure 3 (Switch position data transmission timing diagram). Decoder outputs D1 CLK through D4 CLK at J1-24, 25, 34 and 68 shall remain high. The low on TXDSEL shall enable buffer U15 and disable buffer U18 putting data bits AO, Al, A2, A3, SELECT, INCR, ENTER and DISPLAY per Table I (Input/output bit levels) on Transmitter Buffer Register Lines TBR1, through TBR8, respectively, for transmission. The low on TBRL (J1-30) shall load the data into the UART'S Transmitter Buffer Register and its low to high transition shall shift the data to the UART'S Transmitter Register where it shall be clocked out TRO (J1-28) and CONTROL PANEL to CPU DATA at J1-5 (HI) and J1-6 (LO) per 3.5.5 and Table I.

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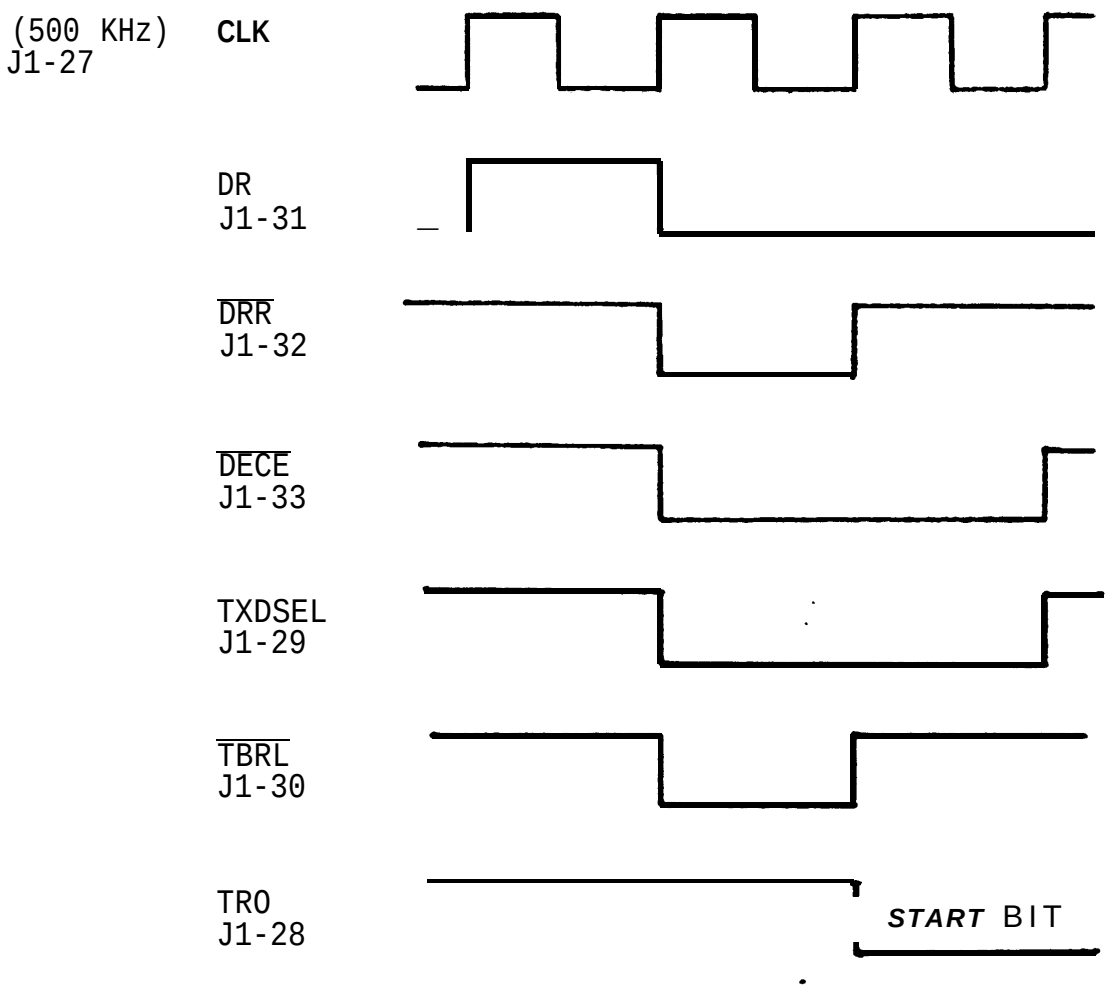


FIGURE 3. Switch position data transmission timing diagram.

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TABLE I. Input/output bit levels for the external Ballistic/BIT Panel.

Input Signal	Input bits at J1-								Output bits at J1-5, 6, TRO							
	10	11	12	13	14	15	16	17	TB R1 LSB	TB R2	TB R3	TB R4	TB R5	TB R6	TB R7	TB R8 MSB
DISPLAY	H	-	-	-	-	-	-	-	-	-	-	-	-	-	-	H
DISPLAY	L	-	-	-	-	-	-	-	-	-	-	-	-	-	-	L
SELECT	-	L	-	-	-	-	-	-	-	-	-	-	L	-	-	-
SELECT	-	H	-	-	-	-	-	-	-	-	-	-	H	-	-	-
INCR	-	-	L	-	-	-	-	-	-	-	-	-	-	L	-	-
INCR	-	-	H	-	-	-	-	-	-	-	-	-	-	H	-	-
ENTER	-	-	-	L	-	-	-	-	-	-	-	-	-	-	L	-
ENTER	-	-	-	H	-	-	-	-	-	-	-	-	-	-	H	-
A0-A3	-	-	-	-	L	L	L	L	L	L	L	L	-	-	-	-
A0-A3	-	-	-	-	H	L	L	L	H	L	L	L	-	-	-	-
A0-A3	-	-	-	-	L	H	L	L	L	H	L	L	-	-	-	-
A0-A3	-	-	-	-	H	H	L	L	H	H	L	L	-	-	-	-
A0-A3	-	-	-	-	L	L	H	L	L	L	H	L	-	-	-	-
A0-A3	-	-	-	-	H	L	H	L	H	L	H	L	-	-	-	-
A0-A3	-	-	-	-	L	H	H	L	L	H	H	L	-	-	-	-
A0-A3	-	-	-	-	H	H	H	L	H	H	H	L	-	-	-	-
A0-A3	-	-	-	-	L	L	L	H	L	L	L	H	-	-	-	-
A0-A3	-	-	-	-	H	L	L	H	H	L	L	H	-	-	-	-
A0-A3	-	-	-	-	L	H	L	H	L	H	L	H	-	-	-	-
A0-A3	-	-	-	-	H	H	L	H	H	H	L	H	-	-	-	-

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3.5.8 End-around-test operation.

3.5.8.1 Initialization. With power and clock applied per 3.5.2 and 3.5.3, respectively, the DCP BIT assembly shall reset per 3.5.6.

3.5.8.2 Data reception and transmission. When a complete input word with format per 3.5.4 and with data bits XXXXX010 or XXXXX101 (DO through D7, respectively) has been clocked into the UART, DR (J1-3U shall transition from low to high. successive clock pulses shall cause the signal transitions shown in Figure 4 (End-around-test timing diagram). Decoder outputs D1 CLK through D4 CLK at J1-24, 25, 34 and 68 and TXDSEL (J1-29) shall all remain high. The high on TXDSEL shall enable buffer U18 and disable buffer U15, wrapping the received data on RBR1 through RBR8 of the UART onto lines TBR1 through TBR8, respectively, of the UART for transmission. The low on TBRL (J1-30) shall load the data into the UART'S Transmitter Buffer Register and its low to high transition shall shift the data to the UART'S Transmitter Register where it shall be clocked out TRO (J1-28) and CONTROL PANEL TO CPU DATA at J1-5 (HI) and J1-6 (LO) with format per 3.5.5.

3.5.9 Display outPut operation.

3.5.9.1 Initialization With power and clock applied per 3.5.2 and 3.5.3, respectively, the DCP BIT assembly shall reset per 3.5.6.

NOTE : The reset operation does not clear the external display to all zeros.

3.5.9.2 Data reception and display control The external FCP CPU sends control words to the DCP BIT assembly that correspond to the Ballistics/BIT Panel FUNCTION SWITCH SETTING. A separate control word for each external display digit must be clocked into the DCP BIT assembly to set/update each digit's value. When a complete control word with format per 3.5.4 and data bits per Table II (Digit 1), Table III (Digit 2), Table IV (Digit 3) or Table V (Digit 4) for the selected digit and value has been clocked into the UART, DR (J1-31) shall transition from low to high. Successive clock pulses shall cause the signal transitions shown in Figure 5 (Display operation timing diagram). The low to high transition of DECE (J1-33) shall disable the 3-to-8 decoder, causing the selected latch's clock input (CLK 1, 2, 3 or 4) to experience a low to high transition. This transition shall cause the lower 5 data bits (DO through D4) to be loaded into the selected latch (latched) and available at that latch's outputs. For digits 1, 2 and 3, Binary Coded Decimal (BCD) to 7-segment decoder/driver ICS shall decode the latched data bits and generate the appropriate output code per Table 11, III or IV for the character to be displayed on the external Ballistics/BIT Panel. For digit 4, the latched data bits (DO through D4) shall address the appropriate location in PROM U24 for the character to be displayed on the external Ballistics/BIT Panel, and the PROM shall output the 7-segment alphanumeric code per Table V corresponding to that address.

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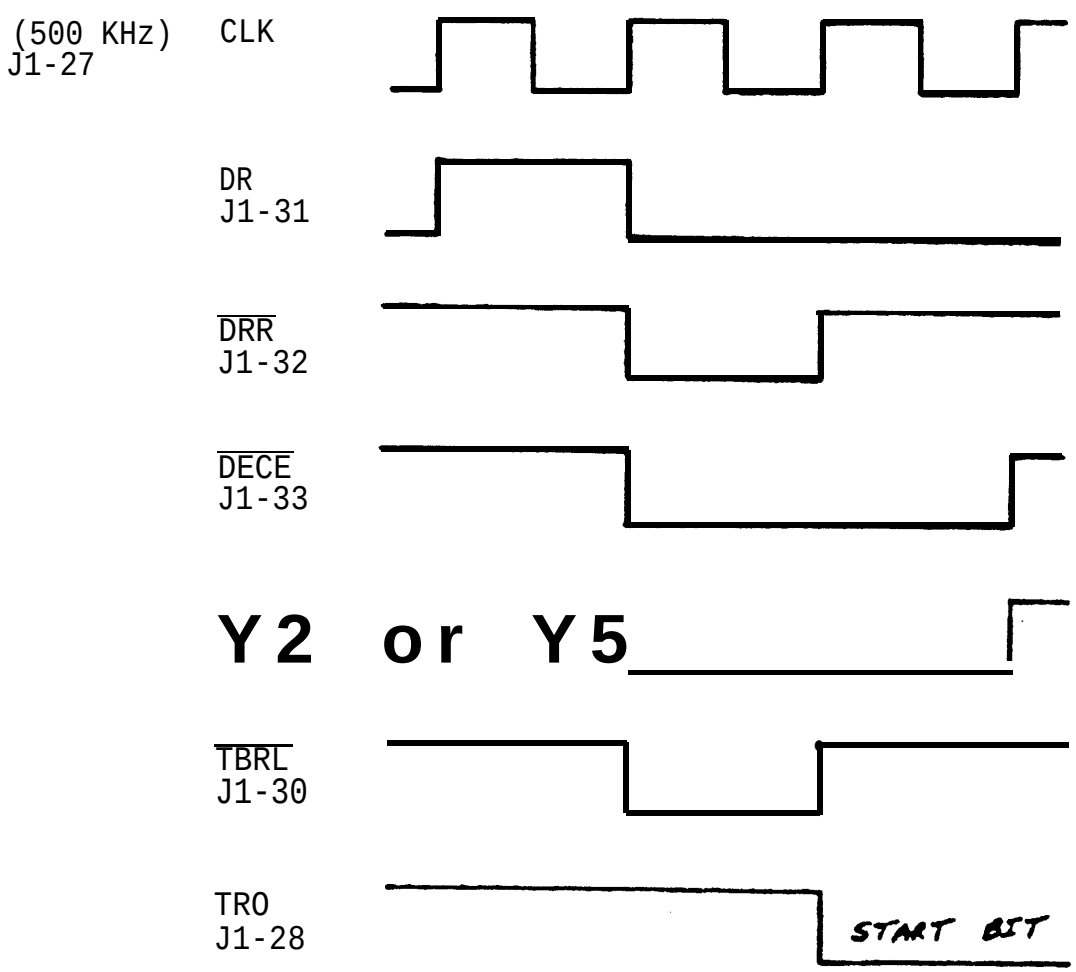


FIGURE 4. End-around-test timing diagram.

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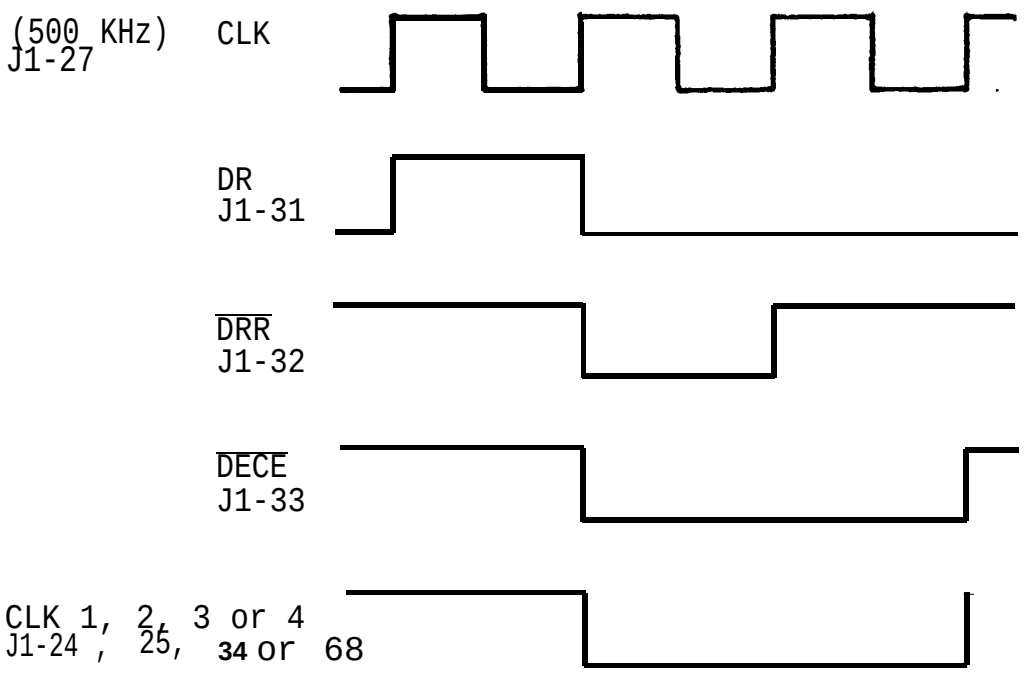


FIGURE 5. Display operation timing diagram.

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TABLE II. Digit 1

Character to be displayed	Input control word at J1-1,2								Output code at Jl-						
	DO	D1	D2	D3	D4	D5	D6	D7	36	37	38	39	40	41	42
0	L	L	L	L	H	L	L	L	L	L	L	L	L	L	H
1	H	L	L	L	X	L	L	L	H	L	L	H	H	H	H
2	L	H	L	L	X	L	L	L	L	L	H	L	L	H	L
3	H	H	L	L	X	L	L	L	L	L	L	L	H	H	L
4	L	L	H	L	X	L	L	L	H	L	L	H	H	L	L
5	H	L	H	L	X	L	L	L	L	H	L	L	H	L	L
6	L	H	H	L	X	L	L	L	H	H	L	L	L	L	L
7	H	H	H	L	X	L	L	L	L	L	L	H	H	H	H
8	L	L	L	H	X	L	L	L	L	L	L	L	L	L	L
9	H	L	L	H	X	L	L	L	L	L	L	H	H	L	L
Blank	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H

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TABLE III. Digit 2

Character to be displayed	Input control word at J1-1,2								Output code at Jl -					
	DO	D1	D2	D3	D4	D5	D6	P7	44	45	46	47	49	50
0	LLL	L	H	Ii	L	L			L	L	L	L	LLH	
1	H	L	L	L	X	H	L	L	H	L	L	H	H	H H
2	L	H	L	L	X	H	L	L	L	LR	L	LH	L	
3	HHL	L	X	H	LL				LL	L	LH	H	L	
4	L	L	H	L	X	H	L	L	H	L	L	H	H	L L
5	H	L	H	L	X	H	L	L	L	H	L	L	H	L L
6	L-H	H	L	X	H	L	L		H	H	L	L	LLL	
7	HEH	L	X	H	L	L			L	L	L	H	H	H H
8	LLLH	X	H	L	L				L	L	L	L	L	L L
9	H	LLH	X	H	LL				L	L	L	H	H	L L
BLANK	L	L	L	L	L	H	L	L	H	H	H	H	H	H H

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TABLE IV. Digit 3

Character to be displayed	Input control word at J1-1,2 D0	D1	D2	1)3	D4	D5	D6	D7	Output code at Jl- 52	53	54	55	56	57	58
0	L	L	L	L	H	H	H	L	L	L	L	L	L	L	H
1	H	L	L	L	X	H	H	L	H	L	L	H	H	HH	
2	L	H	L	L	X	H	H	L	L	L	H	L	L	H	L
3	H	H	L	L	X	H	H	L	L	L	L	L	H	H	L
4	L	L	H	L	X	H	H	L	H	L	L	H	H	L	L
5	H	L	H	L	X	H	H	L	L	H	L	L	H	L	L
6	L	H	H	L	X	H	H	L	H	H	L	L	L	L	L
7	H	H	H	L	X	H	H	L	L	L	L	H	H	H	H
8	L	L	L	H	X	H	H	L	L	L	L	L	L	L	L
9	H	L	L	H	X	H	H	L	L	L	L	H	H	L	L
BLANK	L	L	L	L	L	H	H	L	H	H	H	H	H	E	H	H

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TABLE V. Digit 4

Character to be displayed	Input control word at J1-1,2								Output code at Ji-						
	DO	D1	D2	D3	D4	D5	D6	D7	60	61	62	63	64	65	66
0	L	L	L	L	H	L	L	H	L	L	L	L	L	L	H
1	H	L	L	L	H	L	L	H	H	L	L	H	H	H	H
2	L	H	L	L	H	L	L	H	L	L	H	L	L	H	L
3	H	H	L	L	H	L	L	H	L	L	L	L	H	H	L
4	L	L	H	L	H	L	L	H	H	L	L	H	H	L	L
5	H	L	H	L	H	L	L	H	L	H	L	L	H	L	L
6	L	H	H	L	H	L	L	H	H	H	L	L	L	L	L
7	H	H	H	L	H	L	L	H	L	L	L	H	H	H	H
8	L	L	L	H	H	L	L	H	L	L	L	L	L	L	L
9	H	L	L	H	H	L	L	H	L	L	L	H	H	L	L
BLANK	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H
A	L	H	L	H	H	L	L	H	L	L	L	H	L	L	L
c	H	H	L	H	H	L	L	H	L	H	H	L	L	L	H
E	L	L	H	H	H	L	L	H	L	H	H	L	L	L	L
F	H	L	H	H	H	L	L	H	L	H	H	H	L	L	L
	L	H	H	H	H	L	L	H	H	H	H	H	H	H	L
u	H	H	H	H	H	L	L	H	H	L	L	L	L	L	H
Top & Bottom Segments	H	L	L	L	L	L	L	H	L	H	H	L	H	H	H

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3.5.9.3 **Display current.** Resistor circuits shall help prevent burnout of display filaments when segments are switched on. . segment output line shall be capable of sinking **25** milliamperes maximum.

3.5.10 Lamp test. With power applied per 3.5.2, grounding J1-43 (LAMP TEST) to +5 V RTN shall cause all digit output lines (J1-36 through 42, J1-44 through 50, J1-52 through 58 and J1-60 through 66) is to light all segments of all 4 digits simultaneously to allow detection of burned out lamps.

3.5.11 Input/output requirements. The UART integrated circuit's input/output signals at the test points shall be CMOS levels per 3.5.14. Except for the UART CMOS test points, power input, and RS-422 clock and data inputs and data output, all inputs, outputs and test points shall be at standard TTL voltage levels per **3.5.12.** Connector J1 inputs are listed in Table VI (DCP inputs). Connector J1 outputs are listed in Table VII (DCP outputs).

3.5.12 TTL voltage levels. The TTL voltage levels for inputs, outputs and test points shall be as follows:

	MIN	TYP	MAX
Voltage Output Low (VOL)	---	0.2	0.4
Voltage Output High (vOH)	**2.4**	3.4	
Voltage Input Low (VIL)		---	0.7
Voltage Input High (VIH)	2.0	---	---

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TABLE VI. DCP inputs

Connector-Pin	Signal Name	Destination	Description
J1-1	CPU to Control Panel Data, HI	U2-2	RS-422 input Per 3.5.13
J1-2	CPU to Control Panel Data, LO	U2-1	RS-422 input per 3.5.13
J1-3	Control Panel Clock (500 KHz) ,HI	U2-6	RS-422 input per 305.13
J1-4	Control Panel Clock(500 KHz),LOX	U2-7	RS-422 input per 3.5.13
J1-10	DISPLAY	U15-2, lA1	Data Display BIT I Panel
J1-11	SELECT	U15-4, 1A2	Digit Select BIT I Panel
J1-12	INCR	U15-6, 1A3	Digit Increment I BIT Panel
J1-13	ENTER	u15-8, 1A4	Data Enter BIT Panel
J1-14	AO	u15-11, 2A1	BIT Panel Interface
J1-15	Al	u15-13, 2A2	BIT Panel Interface
J1-16	A2	U15-15, 2A3	BIT Panel [Interface
J1-17	A3	U15-17, 2A4	BIT Panel Interface
J1-18	RESET	C3	BIT Panel Interface
J1-43	LAMP TEST	U9-3, U13-3 U16-3, U24-17	Data Display BIT Panel

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TABLE VII. DCP outputs

Connector-Pin	Signal Name	Source Output	Description
J1-5	Control Panel to CPU Data, HI	U5-2	RS-422 Output per 3.5.13
J1-6	Control Panel to CPU Data, LO	U5-3	RS-422 Output per 3.5.13
J1-36	H	U9-13	Display Digit 1 (LSD)
J1-37	J	U9-12	Display Digit 1 (LSD)
J1-38	K	U9-11	Display Digit 1 (LSD)
J1-39	L	U9-10	Display Digit 1 (LSD)
J1-40	M	U9-9	Display Digit 1 (LSD)
J1-41	N	U9-15	Display Digit 1 (LSD)
J1-42	P	U9-14	Display Digit 1 (LSD)
J1-44	A	U13-13	Display Digit 2
J1-45	B	U13-12	Display Digit 2
J1-46	c	U13-11	Display Digit 2
J1-47	D	U13-10	Display Digit 2
J1-48	E	U13-9	Display Digit 2
J1-49	F	U13-15	Display Digit 2
J1-50	G	U13-14	Display Digit 2
J1-51	S1	U25-4	BIT Panel Interface
J1-52	H	U16-13	Display Digit 3
J1-53	J	U16-12	Display Digit 3

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TABLE VII. DCP outputs (continued)

Connector-Pin	Signal Name	Source Output	Description
J1-54	K	U16-11	Display Digit 3
J1-55	L	U16-10	Display Digit 3
J1-56	M	U16-9	Display Digit 3
J1-57	N	U16-15	Display Digit 3
J1-58	P	U16-14	Display Digit 3
J1-60	A	U20-6	Display Digit 4 (MSD)
J1-61	B	U20-3	Display digit 4 (MSD)
J1-62	C	U20-8	Display Digit 4 (MSD)
J1-63	D	U20-11	Display Digit 4 (MSD)
J1-64	E	U23-6	Display Digit 4 (MSD)
J1-65	F	U23-11	Display Digit 4 (MSD)
J1-66	G	U23-8	Display Digit 4 (MSD)

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3.5.13 RS-422 inputs/outputs.

3.5.13.1 Differential voltage The RS-422 clock, line receiver and line transmitter buffer differential voltage levels shall be as follows:

	High Logic Level	Low Logic Level
Maximum input	+5.5 v	-5.5 v
Sensitivity	$V_{in} = V_{in} - \overline{V_{in}} > +0.5$ v	$V_{11} = V_{in} - \overline{V_{in}} < -0.5$ V

3.5.13.2 Common mode voltages. The RS-422 common mode voltage levels shall be:

	MIN	TYP	MAX
Voltage Output Low (VOL)	---	- -	0.45
Voltage Output High (VOH)	2.5	**3.2**	- -
Voltage Input Low (VIL)	- - -	- - -	0.8
Voltage Input High (VIH)	2.0	- - -	- - -

3.5.13.3 Frequencies. The clock and receiver/transmitter data frequencies shall be:

clock	Data
500.000 KHz ± 74 Hz	31.250 KHz ± 5 Hz

3.5.14 CMOS voltage levels. The CMOS voltage levels for UART test points are as follows:

	MIN	**TYP**	MAX
Voltage Output Low (VOL)	- - -	**---**	0.45
Voltage Output High (VOH)	**2.4**	- - -	- - -
Voltage Input Low (V'IL)	--—	- - -	0.8
Voltage Input High (VIH)	3.5	- - -	- - -

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3.5.15 Test points and spares. The DCP BIT assembly shall have test points and spares as listed in Table VIII (Test points and spares).

TABLE VIII. Test Points.

Connector-Pin	Signal Name	Designation	Description
J1-7	DO	U7-18	Test Point
J1-8	D1	U7-16	Test Point
J1-9	D2	U7-14	Test Point
J1-19	D3	U7-12	Test Point
J1-20	D4	U7-9	Test Point
J1-21	D5	U7-7	Test Point
J1-22	D6	U7-5	Test Point
J1-23	D7	U7-3	Test Point
J1-24	D1 CLK	U6-15	Test Point
J1-25	D2 CLK	U6-14	Test Point
J1-26	RCV	U2-3	Test Point
J1-27	CLK	U2-5	Test Point
J1-28	TRO	U10-25	Test Point
J1-29	TXDSEL	U6-9	Test Point
J1-30	$\overline{\text{TBRL}}$	U11-8	Test Point
J1-31	DR	U10-19	Test Point
J1-32	DRR	U22-6	Test Point
J1-33	$\overline{\text{DECE}}$	U22-9	Test Point
J1-34	D3 CLK	U6-12	Test Point
J1-67	MR	U3-4	Test Point
J1-68	D4 CLK	U6-11	Test Point

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3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3) .

3.7 Soldering. All soldering shall be in accordance with the requirements of Note 2 of drawing 12950877.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance All items must meet all requirements of Sections 3 and 5. The inpections set forth in this specification shall become part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. * The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections The inspection requirements specified herein are classified as follows:

First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c* Special sampling inspection (see 4.5) .

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

	Temperature	+50°F to +90°F
b.	Relative humidity	90% RH max.
c.	Pressure	28 to 32 inches of Hg (700 to 815 mmHg)

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4.3 2 article inspection.

4 .3.1 Sample. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quanities indicated.

Name	Drawing	Quantity
DCP BIT Assemblies	9360884	3

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of any assembly, component or test specimen to comply with any of the requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation The term “inspection lot” is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations’ and tests.

4.4.2.1 Components, assemblies and subassemblieS.

a. The contractor’s quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a

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minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP) . In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

b. Alternative inspection procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. Al 1 approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

4.4.2.2 inspection. Subsequent to first article approval, examinations and tests-related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table IX shall be performed on a 100% basis.

TABLE IX. Functional tests (100 percent inspection).

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Electrical	3.4	4.7.2
102. Performance	3.5	4.7.3
Minor:	None	N/A

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4.5 Special sampling. One DCP BIT assembly out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table IX and shall then meet the tests in Table X. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE X. Special sampling.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Cr Critical:	None	N/A
Major:		
103. Vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105. Altitude	3.3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107. Humidity	3.3.5	4.7.1.5
Minor:	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this Specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment

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is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. contractor inspection equipment design shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table XI (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

TABLE XI. Test equipment list.

ITEM NUMBER		DESCRIPTION
1.	Digital Voltmeter	& 30 Vdc range ± 0.1 Vdc accuracy 20 K ohms input impedance minimum
2.	Digital Card Test	AN/USM-465A
3.	Corn Code Origran Tape	Version 2.0
4.	Test Program Set Manual	A 9361343
5.	Test Program Tape	TPT 9361343
6.	Interface Device Card C	9397373
7.	Test Cable WI	9397501
8.	Test Plug Jll	9361257

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of

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paragraph II-2.1 apply. The DCP BIT assembly shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the DCP BIT assembly shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the DCP BIT assembly shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating). This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

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4.7.3 Performance. The performance of the DCP BIT assembly shall be tested as follows using the equipment listed in Table XI in accordance with the DCP Test Program Set (9361343).

4.7.3.1 UART receiver circuit. The UART receiver circuit shall be tested to meet. the requirements of 3.5.4 in accordance with module 5000 of the TPS 9361343.

4.7.3.2 UART transmitter circuit The UART transmitter circuit shall be tested to meet the requirements of 3.5.5 in accordance with module 6000 of TPS 0361343.

4.7.3.3 Master reset circut. The master reset circuit shall be tested to meet the requirement; of **3.5.6** in accordance with module 3000 of TPS 9361343.

4.7.3.4 switch position transmission. The switch position transmission circuit shall be tested to meet the requirements of 3.5.7 in accordance with module 4000 of TPS 9361343.

4.7.3.5 EAT test. The EAT mode shall be tested to meet the requirements of 3.5.8 in accordance with module 7000 of TPS 9361343.

4.7.3.6 Display output circuit. The display output circuit shall be tested to meet the requirements of 3.5.9 in accordance with module 8000 of TPS 9361343.

4.7.3.7 Lamp test circuit The lamp test circuit shall be tested to meet the requirement; of 3.5.10 in accordance with module 8000 of TPS 9361343.

4.8 soldering Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of note 2 of drawing 12950877 as specified in 3.7.

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5 (PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Special Packaging Instruction PS 9360884. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory)

6.1 Intended use. The DCP BIT assembly interfaces the Ballistics/BIT Panel on the gunner's control panel and the Input/Output Board (IOB) in the Fire Control Processor (FCP) to: transmit the status of the Ballistics/BIT Panel switches to the FCP, conduct an End-Around-Test (EAT) of the Universal Asynchronous Receiver Transmitter (UART) on the DCP BIT assembly, and converting and loading alphanumeric data into the digital display of the Ballistics/BIT Panel. Data transmission is noncontinuous and controlled and serially formatted through a UART on the IOB, which is controlled by the FCP CPU. The DCP BIT assembly is a 4-layered, 5.25 inch by 5.5 inch circuit card mounted horizontally in the AZ control panel. Refer to Figure 6 (DCP BIT assembly functional block diagram) .

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1) .

c. Applicable stock number.

d. First article sample requirements.

e. Packaging requirements, if other than specified in Section 5.

f. Serialization requirements, if applicable.

g. Certification of conformance for each lot of shipment of product.

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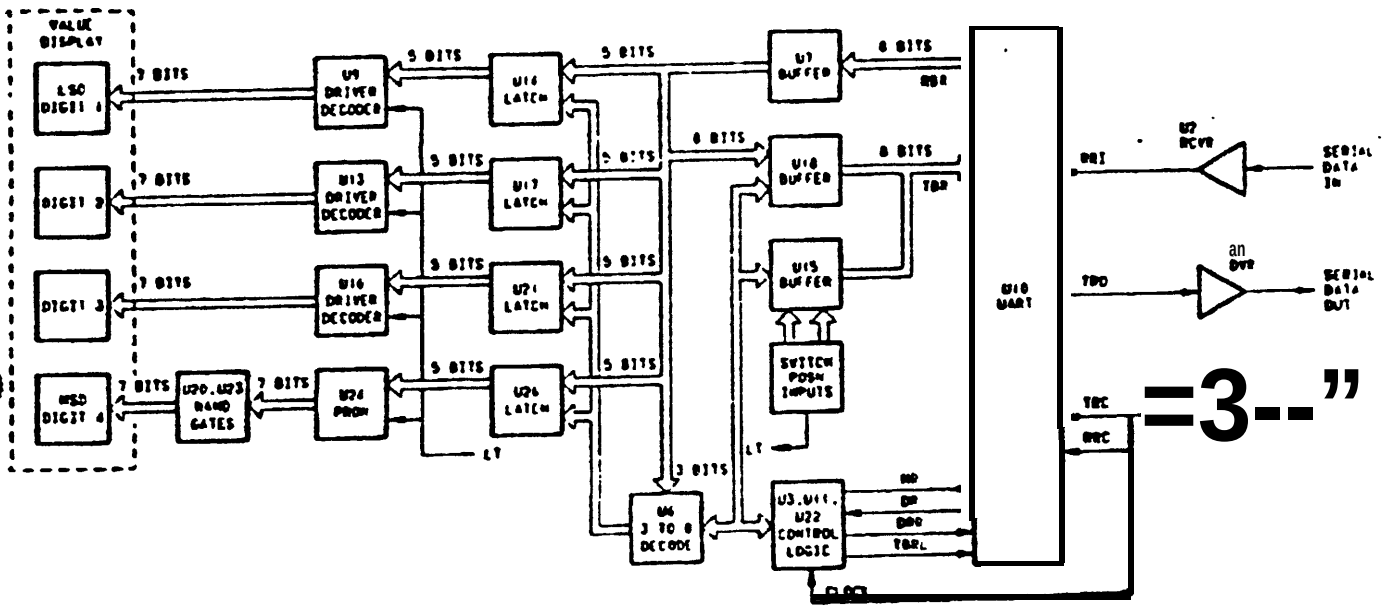


FIGURE 6. Display control panel BIT assembly functional block diagram.

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6.3 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN : SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADC0M), Frankford Arsenal Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Built In Test (BIT)
Fire Control System (FCS)
Vulcan Air Defense System (VADS)

6.6 Changes from Previous issue The previous issues were for ARDEC internal use only. They were not forwarded to Naval Publication and Forms Center (NPFC) for publication. This issue is the initial public release of unlimited distribution.

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